ANDREA ZAMPETTI

PhD student in Environmental and Evolutionary Biology

Click for my profiles:

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CV HIGHLIGHTS

- H-index: 3
- 5 publications on peer-reviewed journals (2 as first author)
- 1 technical report
- Reviewer for 9 scientific journals
- Advanced ecological modelling & AI for wildlife monitoring
- Fieldwork in challenging environments

PERSONAL STATEMENT

I am a PhD student in Environmental and Evolutionary Biology at Sapienza University of Rome, working at the intersection of ecology, conservation biology, and artificial intelligence. My research focuses on developing and applying AI and computer vision approaches for wildlife monitoring, particularly through camera-trapping methods for estimating population density and abundance. I am especially interested in advancing automated data processing pipelines using machine learning algorithms to improve species detection, classification, and density estimation. By integrating ecological modelling with AI-based analytical tools, I aim to enhance large-scale conservation assessments and decision-making processes through accurate, scalable, and data-driven monitoring of wildlife populations.

CURRENT POSITION

Nov 2023 - present — **PhD student in Environmental and Evolutionary Biology**
Sapienza University of Rome, Italy

- Supervisor: Dr. Luca Santini - ECAS lab (*Ecology and Conservation Across Scales*)
- Applying wildlife population density data to conservation studies across different spatial scales through modelling and integrated approaches

EDUCATION

Oct 2022 – Jul 2023 — **MSc in Ecobiology**
Sapienza University of Rome, Italy – 110/110 Cum Laude

- **Thesis**: “Density estimation with camera-traps: comparison between two methods and evaluation of the effects of a machine learning approach for images classification”

Oct 2018 – Oct 2021 — **BSc in Biological Sciences**
Sapienza University of Rome, Italy – 107/110

- **Thesis**: “Aggression and conflict: an overview of mammalian mating systems”

Sep 2013 – Jul 2018 — **Scientific high school diploma**
I.I.S. “Leonardo da Vinci” high school, Maccarese, Italy - 110/110 cum laude

RESEARCH EMPLOYMENTS

Nov 2024 - present

Research contract (junior): "Machine learning applications for automatic species recognition from camera-trap data"

Sapienza University of Rome, Italy

- P.I.: Dr. Luca Santini, Dept. of Biology and Biotechnologies "Charles Darwin"
- Funded by uGov/000301_22_NaturaConnect_HorizonEU-CL6-2021-BIODIV01_CUP_B83C21001860006_SANTINI
- Developing and applying integrated methods for automated species classification and density estimation in camera-traps

Nov 2023 – Nov 2024

Research contract (M2, full-time): AYUDAS RAMÓN Y CAJAL 2021

Museo Nacional de Ciencias Naturales (CSIC), Madrid, Spain

- P.I.: Dr. Ana Benítez-López, Dept. Of Biogeography and Global Change
- Funded by proj. RYC2021-031737-I, MCIN/AEI/10.13039/501100011033, "NextGenerationEU"/PRTR
- Developing a machine learning algorithm for automated classification of arboreal mammal and bird species in the Neotropics

SCIENTIFIC PRODUCTION

Peer-reviewed articles

Giuliani, M., Mirante, D., Russo, L. F., **Zampetti, A.**, & Santini, L. (2026). SANE: An Index of Anthropogenic Noise Levels for Wildlife Research in Terrestrial Ecosystems. *Methods in Ecology and Evolution*. https://doi.org/...

Zampetti, A., Santini, L., Ferreiro-Arias, I., Paltrinieri, L., Ortiz, I., Cedeño-Panchez, B.A., Baltzinger, C., Beirne, C., Bowler, M., Guilbert, E., Kemp, Y.J.M., Peres, C.A., Scabin, A.B., Whitworth, A., & Benítez-López, A. (2026). Introducing TropiCam-AI: A taxonomically flexible automated classifier of Neotropical arboreal mammals and birds from camera-trap data. *Methods in Ecology and Evolution*. https://doi.org/10.1111/2041-210x.70213

Zampetti A., Mirante D., Palencia P., Santini L. (2024). Towards an automated protocol for wildlife density estimation using camera-traps, *Methods in Ecology and Evolution*. https://doi.org/10.1111/2041-210X.14450

Santini L., Mendez Angarita V., Karoulis C., Fundarò D., Pranzini N., Vivaldi C., Zhang T., **Zampetti A.**, Gargano S., Mirante D., Paltrinieri L. (2024). TetraDENSITY 2.0: A database of population density estimates in Tetrapods, *Global Ecology and Biogeography*, e13929. https://doi.org/10.1111/geb.13929

Mirante D., Ancillotto L., **Zampetti A.**, Coiro G., Pisa G., Santocchi C., Giuliani M., Santini L. (2024). Finetuning coexistence: wildlife's short-term responses to dynamic human disturbance patterns, *Global Ecology and Conservation*, e03053. https://doi.org/10.1016/j.gecco.2024.e03053

Technical reports

Santini L., & **Zampetti A**. (2025). National-scale estimates of mammal species abundance in Italy for the Habitat Directive reporting. *Technical Report*. https://doi.org/10.32942/X2V93M

SOFTWARE & ALGORITHMS

AI model

Rodent species classifier from footprint track images

- Creator of the algorithm
- Available on Huggingface at huggingface.co/spaces/andrewzamp/rodent-footprint-classifier

Web-app

TetraDENSITY: A Global Database of Tetrapod Population Density Estimates

- Creator and co-maintainer of the interactive web-app interface
- Available at andrewzamp.github.io/TetraDENSITY

AI model | **TropiCam-AI: A taxonomically flexible automated classifier of Neotropical arboreal mammals and birds from camera-trap data**

- Creator of the algorithm
- Available on AddaxAI at addaxdatascience.com/addaxai

CONFERENCE PRESENTATIONS

Jun 2026 Bolzano, IT

Poster – "Harnessing Artificial Intelligence for small mammals monitoring through non-invasive trapping"
14th Italian Congress of Mammalogy (ATIt)
Best Poster Presentation Award

Jun 2025 Brisbane, AU

Oral – "Introducing *TropiCam-AI*: An automated classifier of Neotropical arboreal mammals and birds from camera-trap images"
ICCB 2025 - 32nd International Congress of Conservation Biology (SCB Oceania)

May 2025 L'Aquila, IT

Oral – "Introducing *TropiCam-AI*: An automated classifier of Neotropical arboreal mammals and birds from camera-trap images"
2nd conference of Conservation Biology for Early Career Researchers (SCB Italy)
Best Oral Presentation Award

Jun 2024 Bologna, IT

Oral – "Towards an automated protocol for wildlife density estimation using camera-traps"
ECCB 2024 - 7th European Congress of Conservation Biology (SCB Italy)

Apr 2023 Rome IT

Poster – "Testing the effect of AI-based species identification on wildlife monitoring"
1st conference of Conservation Biology for Early Career Researchers (SCB Italy)

GRANTS & AWARDS

Dec 2025

"Avvio alla Ricerca" Research Grant (1.200€)
Sapienza University of Rome, Italy
"HAIRS: Hairs AI Recognition Support for small mammals monitoring"

May 2025

Best Poster Presentation Award (300€)
14th Italian Congress of Mammalogy (ATIt)
"Harnessing Artificial Intelligence for small mammals monitoring through non-invasive trapping"

May 2025

Best Oral Presentation Award
2nd conference of Conservation Biology for Early Career Researchers (SCB Italy)
"Introducing TropiCam-AI: An automated classifier of Neotropical arboreal mammals and birds from camera-trap images"

RESEARCH PROJECTS & COMMISSIONS

Nov 2024 – Jun 2025

AI algorithm developer – Rodent species classifier from footprint track images
Sapienza University of Rome, Italy – ECAS lab (Ecology and Conservation Across Scales)

- In collaboration with NEMO (*Nature And Environment Management Operators*)
- Trained AI algorithm for automated classification of rodent species from footprint track images

Nov 2024 – Jun 2025

Ecological modeller – 5th Habitat Directive Reporting
Sapienza University of Rome, Italy – ECAS lab (Ecology and Conservation Across Scales)

- In collaboration with ATIt (*Theriological Italian Association*), University of Turin, and ISPRA (*Superior Institute for Environmental Protection and Research*)
- Contributed as experienced modeller to evaluate mammal population abundance for Italy's 5th Habitat Directive Report

Nov 2023 – Feb 2024 **Field technician – TROPECOLNET project** ("Characterizing the role of interacting species, the drivers and the structure of plant-frugivore ecological networks along a defaunation gradient in tropical forests")
Medio Juruá Extractive Reserve, Amazonas (Brazil)
- P.I.: Dr. Ana Benítez-López
- Fieldwork expedition logistic
- Camera-traps placement, maintenance and data management: visual recognition of mammal and bird species

Mar 2022 – Jul 2023 **Field assistant for a PhD project** ("Impact of ecotourism on species spatial habitat use")
Tenuta Sant'Egidio reserve in Soriano nel Cimino (VT), Italy
- P.I.: Dr. Luca Santini, Dr. Davide Mirante
- Camera-traps placement, maintenance and data management: visual recognition, activity patterns and density estimation for mammal species
- Audio-recording devices placement and maintenance for acoustic surveys of birds and bats

MEDIA & OUTREACH

Books

"The role of Artificial Intelligence in the study of biodiversity" (2026). Featured section in "Conservation Biology", Hoepli Editore, https://www.hoeplieditore.it/universita/articolo/biologia-della-conservazione-luca-santini/9788836014170/2951

Interviews

"Most wildlife AI focuses on the ground. This model looks up in the trees" (2026). Mongabay, https://news.mongabay.com/2026/05/most-wildlife-ai-focuses-on-the-ground-this-model-looks-up-in-the-trees

"Automatic identification of animal species in the Amazon" (2026). Spanish National Radio, https://www.rtve.es/play/audios/a-golpe-de-bit/identificar-automaticamente-especies-animales-amazonia/16951488

Talks

"The virtual ecologist: Artificial Intelligence to study wildlife" (2026). *BioSpritz, Biodiversity Aperitifs* initiative, organized by the Society of Conservation Biology (SCB) – Italy Chapter

TEACHING & SUPERVISION

Teaching

May 2026 Lecture: "*Artificial Intelligence and big-data for large scale camera-trap analysis in wildlife conservation*"
Conservation Biology course, BSc in Biological Sciences, Sapienza University of Rome

Mar 2025 Practical seminar: "*Wildlife occupancy modelling and activity patterns estimation using camera-traps*"
Vertebrate Zoology course, MSc in Ecobiology, Sapienza University of Rome

Mar 2025 Lecture: "*Surveying wildlife through (mostly) non-invasive techniques*"
Conservation Biology course, BSc in Biological Sciences, Sapienza University of Rome

Dec 2024 — Lecture: "*Eyes in the forest: camera-traps for wildlife monitoring and conservation*"
Vertebrate Zoology course, MSc in Ecobiology, Sapienza University of Rome

Students supervision

Sep 2025 – Ongoing — Valerio Ferretti (BSc student in Biological Sciences): "*Drones in wildlife monitoring and conservation: applications and case studies*"

Feb 2025 – Oct 2025 — Matteo D'Alessandro (BSc student in Biological Sciences): "*Density estimation of the crested porcupine (Hystrix cristata) in a private reserve in Central Italy*"

Sep 2024 – Jul 2025 — Gabriele Ferraro (MSc student in Ecobiology; co-supervision): "*Estimates of carnivore densities and habitat use in the Anjozorobe-Angavo protected area, Madagascar*"

Jul 2024 – Oct 2024 — Eleonora Lenci (BSc student in Biological Sciences): "*Optimization of the analysis of camera-trap data in conservation studies: from citizen science to machine learning*"

SCIENTIFIC SOCIETIES

Mar 2026 - ongoing — Society for Conservation Biology (SCB) – Italy Chapter
Member

Feb 2026 - ongoing — Italian Theriological Society (ATIt)
Member

CORE SKILLS

- Excellent written and verbal communication skills
- Quick to learn and adapt to new tools, methods, and environments
- Creative and lateral thinker with strong problem-solving abilities
- Collaborative and team-oriented, fostering productive partnerships
- Ambitious, adaptable, and proactive in achieving project objectives

TECHNICAL COMPETENCY

Coding & Data Science

- R (advanced): statistical modelling, simulations, machine learning, data visualization
- Python/Conda (intermediate): environment management, computer vision training workflows
- AI/ML: model training for image classification (e.g. species detection & identification)
- Integration of AI agentic frameworks for cross-language coding and web apps development

Data Management & Analysis Tools

- GitHub for version control, collaborative coding, and project documentation
- Microsoft Excel, Word, PowerPoint (advanced)
- Supercomputing workflows: SSH/bash scripting for large-scale analyses
- QGIS for spatial analyses

LANGUAGE COMPETENCY

Italian: native
English: written C1 (advanced), comprehension C2 (proficient), spoken C1 (advanced)
Spanish: comprehension B1 (basic)

REFERENCES

Dr. Luca Santini

- Associate Professor, Department of Biology and Biotechnologies "Charles Darwin", Sapienza University of Rome
- Relationship: PhD supervisor
- Email: luca.santini@uniroma1.it
- Tel: (+39) 351 803 1074